# Chemical-aware Wine Purchase Transaction Screening

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## Problem definition and intended use

This project screens wine purchase transactions for unusual purchasing behaviour and inconsistencies between the declared cultivar and a linked laboratory sample. It combines transaction information, including price, quantity and purchase frequency, with a chemical consistency score. Flagged purchases receive a risk score and supporting explanations for human review. Purchase records and anomaly scenarios are explicitly synthetic. A flag does not establish fraud, authenticity, adulteration or food safety.

## Chemical profiling foundation

Milestone 1 established the 13-feature chemical dataset, data quality checks and preprocessing pipeline for cultivar classification. This component provides the chemical consistency signal used in the transaction screening system developed in Milestone 2. The predicted cultivar is compared with the recorded declaration, and the result is attached to a purchase through a sample identifier. The chemical consistency branch therefore depends on the foundation developed in Milestone 1.

The two inputs serve different purposes. Purchase features describe transaction behaviour; chemical measurements support a sample-to-declaration consistency check. A valid sample link and compatible laboratory measurements are required for that check. The system combines both signals to prioritize purchases for review.

## Measured outcome

| Held-out measure | Transaction only | Combined system |
|---|---|---|
| Macro F1 | 0.846 | 0.909 |
| Anomaly recall | 0.590 | 0.940 |
| False alarms / 440 normal purchases | 1 | 26 |
| Missed anomalies / 100 | 41 | 6 |

The gain is conditional on the simulator containing chemical label-mismatch scenarios. These measurements are evidence of prototype behaviour, not estimates of real-world fraud performance.

# Data, provenance and scope

## Real chemical observations; synthetic purchase records

The unchanged Milestone 1 CSV contains 178 UCI Wine samples, 13 numeric chemical measurements and three encoded cultivar classes [1,2]. No missing values or exact duplicate feature vectors were found in Milestone 1. Its 17 descriptive IQR flags were retained. These are laboratory observations, not observed wine transactions. The original class labels are encoded 0/1/2 and do not identify named commercial products.

| Partition | Independent wine samples | Synthetic purchases | Anomaly labels |
|---|---|---|---|
| train | 124 | 2480 | 501 |
| validation | 27 | 540 | 130 |
| test | 27 | 540 | 100 |

## Generator and labels

Twenty purchases are generated for each sample. Scenario probabilities are normal 80%, declaration mismatch 8%, financial anomaly 8% and combined 4%. The binary target is whether a non-normal latent scenario was sampled. It is not calculated by thresholding the classifier score. Mismatch scenarios change the declared cultivar with probability 0.92; other scenarios have incidental declaration changes with probability 0.02. This deliberately creates ambiguity.

Base unit price is lognormal with median 25 and log-scale standard deviation 0.38; quantity is an integer from 1 to 12; recent purchases are Poisson(1); new merchant category occurs with probability 0.15. Financial scenarios perturb price with probability 0.75 by a factor of 0.25 or 3.5, increase quantity with probability 0.70 by 10 to 39, add 2 to 8 recent purchases, and use a new category with probability 0.65. Prices are simulation units, not market observations. Split-specific generator seeds are 100, 101 and 102.

## Risk inputs and excluded information

| Input | Meaning |
|---|---|
| unit_price; quantity | Purchase size and price signals |
| purchases_last_hour; new_merchant_category | Available transaction history aggregates |
| 1 - P(declared cultivar \| chemistry) | Chemical disagreement score |
| predicted cultivar != declared cultivar | Binary laboratory-to-label mismatch |

Sample and transaction IDs only link records. True cultivar, anomaly label and scenario are excluded from risk features. True cultivar is used for chemical supervision and constructing the synthetic scenarios. No merchant identities, personal data or real payment information are present. Real use would require purchase-linked laboratory results; most ordinary retail purchases would not supply them.

# Model selection and architecture

| Family | Fit to the task | Limitations |
|---|---|---|
| Linear / logistic classifier | Compact tabular baseline; fast inference and simple decision surface. Implemented in PyTorch for risk comparison. | Cannot readily model interactions; feature scaling matters. |
| Small multilayer perceptron | Nonlinear interactions among chemical and purchase signals; compact dense numeric input. | Can overfit small samples; scores need calibration; local explanations require extra analysis. |
| Tree-based ensemble | Plausible alternative for tabular thresholds and nonlinear interactions; scaling less critical. | Not evaluated here; extra tuning and larger trees may add memory and complexity. |

## Selected two-stage architecture

Chemical classifier: 13 numeric inputs - 16 hidden ReLU units - 3 logits. Median imputation and z-score scaling are fitted on training samples only; feature order matches Milestone 1. Cross-entropy accepts logits and integer labels [3]. Softmax is applied only when obtaining probabilities. The hidden layer expresses a compact nonlinear mapping without assuming sequence, spatial or graph structure.

Risk classifier: 6 standardized inputs - 16 hidden ReLU units - 2 logits. It uses four purchase fields plus two chemical consistency features. The small MLP represents interactions such as an unusual purchase accompanied by a declaration mismatch. Its combined validation macro F1 was 0.9721, narrowly above the weighted linear alternative. The architecture was frozen before test evaluation. The linear alternative later performed better on test, which is reported rather than used to reselect the model.

## Versioned training configuration

| Setting | Value |
|---|---|
| Configuration / random seed | config version 2.0 / seed 42 |
| Optimizer / learning rate / regularization | Adam / 0.01 / weight decay 0.001 |
| Batch size / maximum epochs | 32 / chemical 250, risk 200 |
| Stopping / restoration | Validation cross entropy; patience 25; minimum improvement 0.00001; restore best checkpoint |
| Imbalance / augmentation | Risk loss uses inverse-frequency class weights; no resampling or augmentation. Chemical loss unweighted. |
| Chemical width candidates | 8, 16, 32; training-only stratified 3-fold comparison |

The two networks contain 421 trainable parameters in total. Large Transformers and CNNs do not have a useful inductive bias for this tiny, unordered numeric table. The comparison is deliberately small to limit repeated validation-set tuning.

# Training protocol and leakage controls

## Preserved Milestone 1 sample assignments

The original stratified seed-42 split is preserved: first hold out 30%, then divide that remainder equally using the same seed. This produces 124 training, 27 validation and 27 test wine samples. All 20 transactions associated with a sample stay in its partition. Neither duplicate purchases nor the same chemical sample can cross the train/test boundary. Dates and batch metadata are unavailable, so a credible temporal or producer holdout cannot be claimed.

## Chemical model development

| Hidden width | Fold 1 F1 | Fold 2 F1 | Fold 3 F1 | Mean F1 |
|---|---|---|---|---|
| 8 | 0.975 | 0.975 | 0.951 | 0.967 |
| 16 | 0.975 | 1.000 | 0.975 | 0.984 |
| 32 | 0.975 | 1.000 | 0.953 | 0.976 |

Three stratified folds of the 124 training samples compare widths. Imputation and standardization are refitted within each fold. Each comparison uses its held-out fold for early stopping as well as scoring, so these are development scores rather than unbiased external estimates. Width 16 is selected. The final chemical model trains on the original training samples and uses the original validation samples for stopping.

## Stacked predictions for risk training

The risk model never trains on chemical predictions obtained by fitting that chemical model to the same sample. Three-fold out-of-fold predictions are generated for training samples. Each fold model uses a further stratified 80/20 split of its fitting portion for early stopping; the outer held-out sample labels do not determine stopping. The globally selected chemical width is reused, so model-selection dependence within the training set remains. Validation and test purchases use the final chemical model fitted on the original training samples.

## Risk comparisons and threshold selection

Weighted transaction-only MLP, weighted combined linear model and weighted combined MLP are trained. An unweighted combined MLP is a separate imbalance ablation. The main architecture comparison is predefined between the two weighted combined candidates; the highest validation macro F1 wins. For each model, 33 thresholds from 0.10 through 0.90 are compared on validation macro F1. The selected combined threshold is 0.45. No test label selects an architecture, stopping epoch or threshold.

The unweighted ablation has slightly higher validation macro F1 than the weighted main model, and identical test decisions after separate threshold tuning. Consequently weighting is not shown to improve this experiment. It is retained as a predefined main-run mitigation, not presented as a proven gain. A future experiment should compare calibration and review cost when choosing weighting.

Hardware: Intel Xeon Platinum 8573C CPU, one PyTorch compute thread, no GPU. The environment reports 9 logical CPUs and 9.73 GiB RAM. Python 3.12.14, PyTorch 2.14.1+cpu and scikit-learn 1.8.0. No epochs are run on validation/test examples as training data.

# Validation and test results

| Risk model | Val macro F1 | Test accuracy | Test macro F1 | Test PR-AUC |
|---|---|---|---|---|
| transaction only | 0.839 | 0.922 | 0.846 | 0.760 |
| combined linear | 0.972 | 0.948 | 0.919 | 0.927 |
| combined mlp | 0.972 | 0.941 | 0.909 | 0.958 |
| combined unweighted | 0.977 | 0.941 | 0.909 | 0.954 |

The combined MLP remains the selected model despite the linear model achieving higher held-out macro F1. Model selection follows validation; test results are not used to switch models. The always-normal baseline attains 440/540 = 0.815 test accuracy but zero anomaly recall, showing why accuracy alone is insufficient.

| Selected model metric | Validation | Test |
|---|---|---|
| Accuracy | 0.980 | 0.941 |
| Macro precision | 0.973 | 0.885 |
| Macro recall / balanced accuracy | 0.971 | 0.940 |
| Macro F1 | 0.972 | 0.909 |
| Micro F1 | 0.980 | 0.941 |
| Anomaly precision | 0.961 | 0.783 |
| Anomaly recall | 0.954 | 0.940 |
| Anomaly F1 | 0.958 | 0.855 |
| AUROC | 0.987 | 0.972 |
| PR-AUC (average precision) | 0.979 | 0.958 |

## Errors and the chemical component contribution

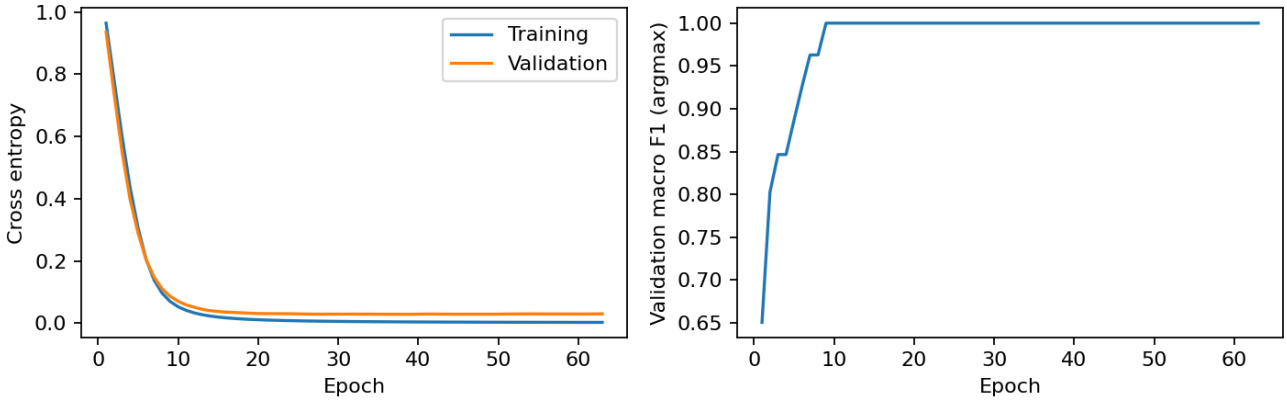
Test confusion matrix, rows actual normal/anomalous and columns predicted normal/anomalous: [[414, 26], [6, 94]]. False-positive rate is 26/440 = 5.91%; 6/100 anomalies are missed. Transaction-only errors are [[439, 1], [41, 59]]. Adding chemistry raises review burden but recovers many declaration mismatches. Label-mismatch-only recall rises from 1/40 to 35/40; financial-only recall is 36/37 for both models. Combined scenarios are flagged 23/23 versus 22/23.

Paired sample-cluster bootstrap: the macro F1 gain is 0.063, with exploratory 95% percentile interval [-0.045, 0.143] from 1,000 resamples of the 27 held-out wine samples. This accounts for shared sample identity, but not generator uncertainty, hyperparameter selection or external distribution shift.

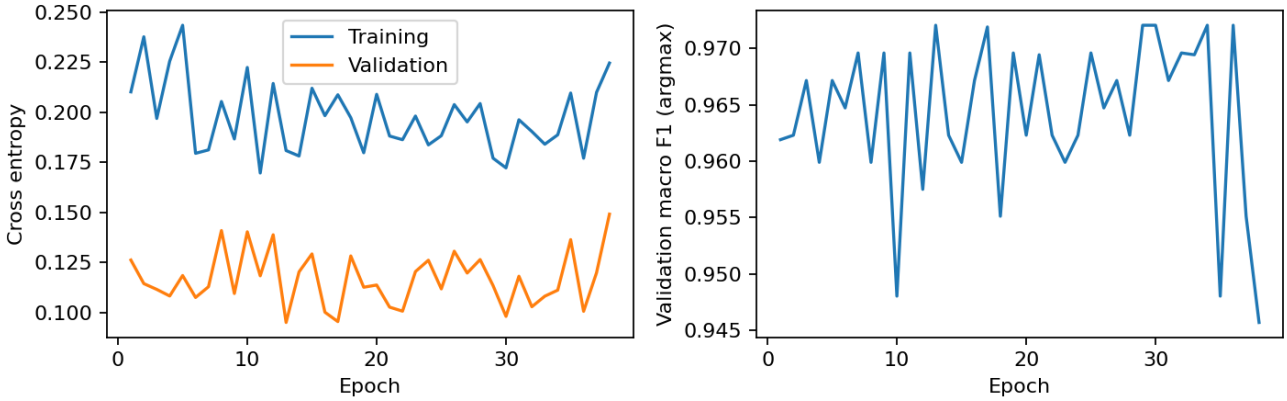
The chemical classifier itself achieves validation macro F1 1.000 and test macro F1 0.967 (26/27 correct), versus majority-class test accuracy 0.407. One changed test prediction moves chemical accuracy by 3.7 percentage points. AUROC and macro average precision are 1.000 despite one argmax error because ranking and classification decisions measure different properties.

# Learning curves and local explanations

## Chemical classifier




## Selected combined risk classifier




Curves include every executed epoch, including patience epochs after the best checkpoint. The chemical model restores epoch 38 of 63; the risk model restores epoch 13 of 38. The F1 curves use argmax decisions, whereas the final risk metrics use the validation-selected 0.45 threshold. CSV histories and plots for all comparison models are included.

## Why each transaction was flagged

For each prediction, the explanation procedure replaces one standardized purchase field, or both chemical consistency fields together, with their training means. The reduction in risk score measures the local support from that group. The three largest score reductions are saved for every test purchase, including all 120 flagged purchases. Negative reductions are retained when a factor lowers rather than supports risk. The demo also reports the predicted versus declared cultivar and descriptive purchase deviations.

These are model sensitivity diagnostics, not causal explanations. Mean substitutions may describe unrealistic records and correlated inputs can redistribute apparent importance. A class mismatch may reflect a laboratory error, incorrect declaration, model error or an unfamiliar cultivar. It does not prove wrongdoing. results/test_explanations.json contains the complete auditable output.

# Training and inference efficiency

| Stage | Executed / best epoch | Training seconds | Seconds / epoch |
|---|---|---|---|
| Chemical classifier | 63 / 38 | 0.594 | 0.009 |
| transaction only | 28 / 3 | 1.692 | 0.060 |
| combined linear | 45 / 20 | 1.624 | 0.036 |
| combined mlp | 38 / 13 | 2.103 | 0.055 |
| combined unweighted | 30 / 5 | 1.811 | 0.060 |

The complete experiment, including width comparisons, out-of-fold chemical models, risk comparisons, benchmarking and plots, took 20.97 seconds. Per-model timing includes validation and history calculation as well as gradient updates; it is not kernel-only compute time.

| Memory / size / arithmetic | Observed value or defined estimate |
|---|---|
| Peak sampled process RAM | 470.2 MiB, sampled every 5 ms across the complete experiment |
| OS process high-water RAM | 469.7 MiB; minor accounting differences from the sampler |
| Inference-stage process RAM | 448.4 MiB; includes Python, libraries and retained training objects |
| Model parameters / checkpoint | 421 parameters / 6185 bytes; raw float32 weights are 1,684 bytes |
| Forward arithmetic estimate | 768 dense FLOPs per purchase; 2 operations per multiply-accumulate |
| VRAM / energy / monetary cost | No GPU used; energy and cost not measured |

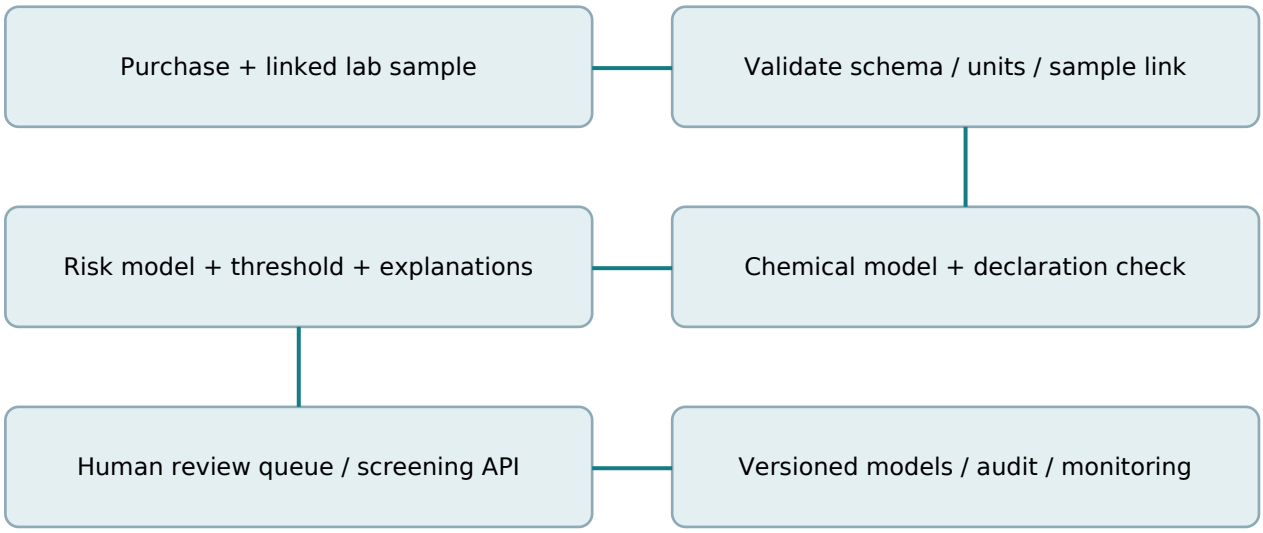
The forward estimate is 2 x (13 x 16 + 16 x 3 + 6 x 16 + 16 x 2) = 768 operations (0.000000768 GFLOPs per purchase). It excludes biases, ReLU, softmax and preprocessing. It is a mathematical count, not measured hardware FLOP/s; training backward-pass FLOPs are not profiled. Low operation count does not imply low Python runtime memory.

| CPU batch | p50 latency ms | p90 ms | p95 ms | Samples / second |
|---|---|---|---|---|
| 1 | 0.541 | 1.271 | 1.642 | 1,162 |
| 32 | 0.613 | 1.418 | 1.959 | 35,093 |

One hundred warm-up calls precede 1,000 timed calls for each batch size. Timed work includes chemical imputation/scaling, both neural networks, chemical consistency calculation, risk scaling and probability output. Batch-32 figures are total batch latency; batch-1 figures are per purchase. Throughput is batch size divided by mean call duration. File loading, JSON serialization, explanation generation, networking and laboratory processing are excluded. Shared hardware causes timing variation.

Quantization, distillation and ONNX export were not attempted; no speedup or accuracy effect is claimed. With only 421 parameters, runtime and data acquisition are likely larger costs than weight storage. A fresh serving-process memory profile is still required before choosing an edge device.

# Deployment architecture and operations




## Online versus offline

Offline jobs fit preprocessing, train chemical and risk models, select thresholds, validate candidates and publish a versioned checkpoint. Laboratory measurement is external and may take much longer than inference. Online screening starts only when the purchase and compatible laboratory record are available. Validate the declared class, feature order, source-scale compatibility, finiteness and linkage before scoring. Chemical inference may be precomputed per sample.

## Hybrid serving and workload

A laboratory workstation or small CPU service computes chemical probabilities near the measurements. A central transaction service joins those probabilities to purchases and manages review/audit records. Local chemical processing reduces repeated transfers and permits limited offline work; central purchase history supports the frequency feature. An edge-only design cannot reliably see cross-device activity. Cloud-only operation simplifies updates but adds connectivity and data-governance dependencies.

Use request/response scoring for a purchase with all required inputs. Use asynchronous queues for delayed laboratory results, explanations and analyst review. A pending lab result returns pending/manual review, rather than an invented chemical score. Offline reconciliation uses microbatches of up to 32 with a bounded waiting time; streaming calls use batch 1. The measured benchmark supports this design but does not exercise a deployed server.

## Cache, failure and rollback

Cache chemical probabilities using sample ID, laboratory-record version and chemical-model version. Invalidate after measurement corrections or model changes. Use transaction IDs as idempotency keys to prevent duplicate review tasks. Keep failed jobs in a retry/dead-letter queue. Missing or incompatible chemistry routes to manual review; an optional transaction-only fallback must be explicitly labelled as reduced coverage. Never silently substitute chemical inputs.

Deploy a model/preprocessor/threshold bundle atomically. Retain the previous validated version and roll back on schema errors, unexplained flag-rate spikes or latency failures. During outage, queue transactions for review or apply the approved transaction-only fallback; the system must not automatically declare the purchase genuine or reject it as fraud.

# Trade-offs, monitoring and reproducibility

## Proposed service objectives, distinct from observed results

| Proposed objective | Evidence and boundary |
|---|---|
| Warm CPU scoring p95 <= 10 ms | Observed batch-1 p95 1.64 ms; excludes explanations, network and laboratory turnaround. |
| API p95 <= 100 ms after inputs arrive | Design target only; no API service or load test is implemented. |
| Offline macro F1 >= 0.85; anomaly recall >= 0.90 | Synthetic test: 0.909 macro F1 and 0.940 recall. Real deployment gate requires independently verified records. |
| False-positive rate <= 0.03 | Not met: 0.059. Investigate calibration, threshold and analyst capacity before deployment. |

Chemistry reduces missed declaration inconsistencies but increases false alarms and requires expensive laboratory input. Lower risk thresholds generally favour recall and increase review volume; higher thresholds can miss anomalies. Thresholds must be tuned with validation outcomes and explicit costs, never adjusted to improve reported test scores. Adding a larger network is not justified by the tiny independent sample count.

## Shadow deployment and monitoring

Run shadow screening alongside existing review for a proposed two-week pilot without automatically blocking purchases. Record score, model version, data completeness, queue delay and analyst disposition. Monitor per-stage p50/p90/p95 latency, failure rates, flag rates, missing lab links, input distributions, predicted cultivar mix and review backlog. With delayed verified labels, monitor precision, recall and false-positive rate by scenario/merchant cohort where sample sizes permit. Compare model scores with investigation outcomes; a changed feature distribution alone does not prove fraud drift.

## Reproduce and inspect

Run experiment.py, explain_results.py, verify.py and build_report.py with Python. Demo.ipynb and demo.py demonstrate saved-model inference. No API keys are needed. Versioned configuration, dependencies, CSV predictions, fold summaries, learning histories, sample partitions and source checksum are included. Verification passes split separation, checkpoint inference, invalid-input rejection and metric recomputation.

Project repository: https://github.com/kamzyy786/Wine-Transaction-Anomaly-Detection. The Milestone 2 code, trained checkpoint, data, report and measured outputs are available in the milestone2 folder. The original Milestone 1 preprocessing files remain at the repository root. Instructor/POD access has not been independently verified.

## References

[1] Aeberhard, S. and Forina, M. Wine [Dataset]. UCI Machine Learning Repository. DOI: 10.24432/C5PC7J. CC BY 4.0. https://archive.ics.uci.edu/dataset/109/wine

[2] scikit-learn. load_wine documentation. https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_wine.html

[3] PyTorch. CrossEntropyLoss documentation. https://docs.pytorch.org/docs/stable/generated/torch.nn.CrossEntropyLoss.html

# Instruction audit and demo readiness

| Instruction / marks | Evidence in this report and package |
|---|---|
| 1. Model selection / 2.5 | Three plausible families with benefits and limits; task match and inductive bias; selected architecture, preprocessing and versioned configuration. Pages 3-4; config.json. |
| 2. Training and experimentation / 3.0 | Preserved stratified sample splits; small grid comparison; hardware and stopping; learning curves; chemistry and loss-weight ablations; validation/test accuracy, precision, recall, macro/micro F1, AUROC and PR-AUC. Pages 4-6; CSV histories and metrics.json. |
| 3. Training efficiency / 1.0 | Total and per-model times, epoch times, observed peak process RAM, parameters/checkpoint size, inference arithmetic and training GFLOPs estimate below. Page 7; efficiency_audit.json. |
| 3. Inference efficiency / 1.0 | CPU and batches 1/32; p50/p90/p95 latency; throughput; fresh-process RAM below. Quantization/distillation/ONNX not attempted and no effects claimed. Page 7. |
| 4. Deployment and trade-offs / 2.5 | Online/offline, hybrid edge/cloud and batch/streaming; ingestion, preprocessing, models and review outputs; cache/queues; fallback/rollback; SLOs and shadow monitoring. Pages 8-9. |
| Submission format and repository | PDF with name and student ID 40325656; Python/PyTorch code, data, checkpoint and results uploaded to the project repository. Instructor/POD access remains unverified. Flag explanations are illustrated on page 11. |

## Additional efficiency evidence

A separate process that loads only the serving code and checkpoint measured 355.0 MiB RAM after imports, 361.0 MiB after checkpoint load, and 367.7 MiB after warm inference. The load increment is 6.0 MiB, not an isolated weight-memory measurement. No training objects are retained. This replaces the need to infer serving footprint from the training process.

The direct-array demo path, including grouped explanation generation, measured batch-1 p50/p90/p95 of 0.266/0.656/0.859 ms after 100 warm-ups over 1,000 calls. It avoids the sklearn transform wrappers used in the earlier pipeline benchmark; runs occur under different shared-host load, so the two timings are not a controlled explanation-overhead comparison.

Dense training compute is estimated as three times dense forward FLOPs per sample (forward plus approximately two forward-equivalents for backward), multiplied by training samples and all executed epochs: chemical model 0.011999 GFLOPs; selected risk model 0.072376 GFLOPs; total 0.084376 GFLOPs. These are analytical estimates, not profiler measurements. Activation, loss, optimizer, validation and CV/OOF computations are excluded.

## What is working and what comes next

The saved-model notebook and command-line demo run now. Verification checks input rejection, preserved splits, sample separation, score agreement and metric recomputation. The deployment diagram is a design: a hosted API, shared transaction history, cache/queue infrastructure and analyst workflow remain future implementation. DEMO_GUIDE.md provides the presentation sequence and the continuation plan. No full production deployment is claimed.

# Flagged purchases: worked examples

The following records are selected from the saved held-out synthetic test predictions. Their scenario labels come from the generator and are shown for evaluation; they are not model inputs. Both exceed the validation-selected decision threshold of 0.45 and are routed to human review.

## Example 1: chemical declaration inconsistency

| Field | Recorded output |
|---|---|
| Synthetic transaction | SYN-102-7-8 |
| Scenario / decision | label mismatch / human review |
| Risk score / threshold | 0.863115 / 0.45 |
| Predicted / declared cultivar | 0 / 1 |

| Feature group | Score reduction at training mean |
|---|---|
| chemical consistency | 0.800216 |
| purchases last hour | 0.074760 |
| quantity | 0.010212 |

Replacing the chemical consistency group with its overall training means reduces this score by about 0.800. The model predicts class 0 while the purchase declares class 1. This supports investigating the sample link, recorded declaration and laboratory measurements; it is not proof of fraud.

## Example 2: unusual purchase behaviour

| Field | Recorded output |
|---|---|
| Synthetic transaction | SYN-102-46-16 |
| Scenario / decision | financial / human review |
| Risk score / threshold | 0.999889 / 0.45 |
| Predicted / declared cultivar | 0 / 0 |

| Feature group | Score reduction at training mean |
|---|---|
| purchases last hour | 0.282335 |
| quantity | 0.007383 |
| unit price | -0.000006 |

The predicted and declared cultivars agree, yet the purchase is flagged. Recent purchase frequency is the largest reported supporting group. The near-zero negative unit-price value indicates that this factor does not positively support the flag in this local diagnostic. The transaction branch can therefore trigger review even without a chemical disagreement.

Occlusion uses overall training means, not normal-only values. Score differences are local sensitivities, not causal effects; correlated inputs and unrealistic substitutions limit interpretation.